MIL-I-55633(EL)
31 October 1968

SUPERSEDING
SC-A-46571B
15 December 1964

MILITARY SPECIFICATION
INDICATOR, PANORAMIC IP-173( )/U

1. SCOPE.

1.1 This specification covers one type of automatic scanning superheterodyne receiver designated as Panoramic Indicator IP-173( )/U. The indicator permits visual analysis and identification of rf signals up to 1 GHz.

2. APPLICABLE DOCUMENTS.

2.1 Documents. The following documents of the issue in effect at date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-P-11268	Parts, Materials and Processes Used in Electronic Communication Equipment.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-I-55591	Indicator Panoramic IP-173( )/U Packaging and Packing of.

FSC 6625

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of.
MIL-STD-781	Reliability Tests Exponential Distribution.
MIL-STD-810	Military Standard Environmental Test Methods for Aerospace and Ground Equipment.

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Tester.
SM-C-340469	Cable Assembly, Spec. Purpose, Electrical.
SM-C-340470	Cable Assembly, Radio Frequency.
SM-C-340561	Constant Voltage Transformer.
SC-DL-387101	Indicator, Panoramic.
SC-DL-387102	Power Supply.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS.

3.1 Description. The Panoramic Indicator IP-173( )/U is an automatic scanning superheterodyne receiver which permits visual analysis and identification of rf signals up to 1 GHz. All signals within a scanned band of up to 1 MHz are displayed simultaneously on the cathode ray tube. The received signal appears as a pip on the cathode ray tube. The pip amplitude and its position along the calibrated horizontal axis of the cathode ray tube screen indicates signal level and frequency. The equipment consists of an indicator, a power supply, a constant voltage transformer, and associated cables. The constant voltage transformer is used to provide a steady 117 Vac, 60 Hz, single phase input to the power supply.

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3.2 First article samples. The contractor shall furnish three (3) complete first article samples for approval, provided the invitation for bid and the contract requires first article samples.

3.3 Construction. The equipment shall be constructed in accordance with the following drawings and the requirements specified herein:

SM-C-340469	Cable Assembly, Spec. Purpose, Electrical.
SM-C-340470	Cable Assembly, Radio Frequency.
SM-C-340561	Constant Voltage Transformer.
SC-DL-387101	Indicator, Panoramic.
SC-DL-387102	Power Supply.

3.4 Parts, materials and processes-general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection of requirements therein, shall apply.

3.5 Finish, protective. The equipment shall be finished in accordance with MIL-F-14072. This includes finish of hardware; such as screws, etc., and necessary touch-up after mounting. The final paint film on type I surfaces shall be final Film E, semi-gloss, light gray enamel, conforming to MIL-F-14072.

3.6 Marking. Marking shall conform to MIL-M-13231.

3.6.1 Visibility. Wherever practicable, parts and assemblies shall be so mounted that their identification markings shall be readily visible with a minimum disassembly of the equipment.

3.6.2 Serial numbers. Serial numbers are required on each nomenclatured and type designated item, except cords and cables.

3.7 Electrical.

3.7.1 Operating voltage. The Panoramic Indicator IP-173( )/U shall operate from a primary power source between 95 Vac and 135 Vac, 60 Hz, single phase to the constant voltage transformer with an output of 117 Vac ± 1.1 Vac when tested in accordance with 4.7.1.

3.7.2 Line positioning.

3.7.2.1 Horizontal. With the horizontal position control at its maximum counter clockwise position, the pip shall be at least two divisions to the left of the center vertical grid line when tested in accordance with 4.7.2.1. With the horizontal position control fully clockwise, the pip shall be at least two divisions to the right of the center grid line when tested in accordance with 4.7.2.1.

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3.7.2.2 Vertical. When the vertical position control is varied over its entire range, the baseline shall vary at least a half inch above and below the zero reference line of the linear scale when tested in accordance with 4.7.2.2.

3.7.3 Sweep rates. The sweep rate selector switch shall select sweep rates of 1, 5, and 30 Hz and a variable sweep rate. The variable sweep rate control shall adjust the sweep rate over the range of 25 to 35 Hz.

3.7.3.1 1 Hz. The average time for twenty sweeps shall be 20 seconds ± 1 second, when tested in accordance with 4.7.3.1.

3.7.3.2 5 Hz. Twelve stationary sine waves shall appear on the screen when tested in accordance with 4.7.3.2.

3.7.3.3 30 Hz. Two stationary sine waves shall appear on the cathode ray tube screen when tested in accordance with 4.7.3.3.

3.7.3.4 Variable sweep rate. With the variable sweep rate control fully counter clockwise, when one stable signal appears, the modulating signal of the signal generator shall be 25 Hz, or less. With the variable sweep rate control at its midpoint, when one stable signal appears, the modulating signal shall be approximately 30 Hz. With the variable sweep rate control fully clockwise, the modulating signal shall be 35 Hz or greater. The equipment shall be tested in accordance with 4.7.3.4.

3.7.4 Hum. With no signal, the sweep rate selector switch at "5", and the gain control fully counter clockwise, the amplitude of any residual signals seen on the baseline shall not exceed 1/64 inch when tested in accordance with 4.7.4.

3.7.5 Sensitivity.

3.7.5.1 Cross product sensitivity. A full scale pip, or larger shall appear on the screen when tested in accordance with 4.7.5.1.

3.7.5.2 Direct input sensitivity. The input shall not exceed 500 microvolts in either test case to produce a full scale deflection when tested in accordance with 4.7.5.2.

3.7.6 Cross product reserve gain. The output of signal generator B shall be 0.03 volt or less when tested in accordance with 4.7.6.

3.7.7 Calibration.

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3.7.7.1 LIN amplitude scale. At each step of signal level, the signal height shall be at or within 1/5 of a division of the LIN calibration line on the screen when tested in accordance with 4.7.7.1.

3.7.7.2 LOG scale. At each step of attenuation, the signal level shall be within ± 1 dB of the corresponding dot on the left side of the screen when tested in accordance with 4.7.7.2.

3.7.7.3 Frequency. Each pip shall fall within 1/5 of a division of its respective grid line, when the pip is initially placed on the center grid line at a frequency of 5 MHz when tested in accordance with 4.7.7.3.

3.7.8 Sweep width.

3.7.8.1 Width. The sweep width range shall be 1000 kHz ± 40 kHz when tested in accordance with 4.7.8.1.

3.7.8.2 Linearity. Each pip shall fall within 1/5 of a division of its respective grid line when tested in accordance with 4.7.8.2.

3.7.9 Centering ranges.

3.7.9.1 Left. A pip shall appear at least 1/4 of a division to the left of the center zero vertical grid line when tested in accordance with 4.7.9.1.

3.7.9.2 Right. A pip shall appear at least 1/4 of a division to the right of the center zero vertical grid line when tested in accordance with 4.7.9.2.

3.7.10 Response curve. The response over an input signal range of 4.5 to 5.5 MHz shall not vary more than 7 percent of full scale when tested in accordance with 4.7.10.

3.7.11 Center frequency image rejection. The center frequency image rejection ratio shall be a minimum of 150.1 when tested in accordance with 4.7.11.

3.7.12 FM operations.

3.7.12.1 Multivibrator (2 Hz). A two hertz sine wave with a superimposed pulse shall appear on the oscilloscope screen when tested in accordance with 4.7.12.1.

3.7.12.2 Multivibrator (3Hz). A three hertz sine wave with a superimposed pulse shall appear on the oscilloscope screen when tested in accordance with 4.7.12.2.

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3.7.12.3 Distortion. The peak-to-peak ripple amplitude shall not exceed three screen divisions when tested in accordance with 4.7.12.3.

3.7.12.4 Intensifier. When the intensifier control is turned clockwise, the signal pip brightness shall increase when tested in accordance with 4.7.12.4.

3.7.13 Resolution capability. The resolution capability at 200 Hz shall be at least 3 dB down, when tested in accordance with 4.7.13.

3.7.14 Drop off at 30 Hz. The drop off at 30 Hz shall not be greater than 20 percent when tested in accordance with 4.7.14.

3.7.15 Video filter operation. The pip shall decrease at least 20 percent of full scale amplitude when tested in accordance with 4.7.15.

3.8 Service conditions. The Panoramic Indicator IP-173( )/U shall be constructed so that no fixed part shall become loose, cracked, flaked, pitted or blistered, no movable part or control shall be shifted in setting, position or adjustment, and no permanent degradation shall be caused in performance below that specified herein when subjected to the following conditions.

3.8.1 Fungus. The equipment shall show no evidence of viable fungus and shall meet all Group A requirements after being tested in accordance with 4.8.1.

3.8.2 Altitude. The equipment shall be subjected to the altitudes specified and shall meet the electrical requirements of 3.7.3 at times specified, when tested in accordance with 4.8.2.

3.8.3 Low temperature. The equipment shall be subjected to the low temperature extremes of -4°F operating and -65°F storage and shall meet the electrical requirements of 3.7.3 at times specified, when tested in accordance with 4.8.3.

3.8.4 High temperature. The equipment shall be subjected to the high temperature extremes of 125°F operating and 160°F storage and shall meet the electrical requirements of 3.7.3 at times specified, when tested in accordance with 4.8.4.

3.8.5 Humidity. The equipment shall exhibit no physical damage of parts and materiel, such as corrosion, rust, blistering, swelling, deterioration etc, or electrical degradation, and shall meet the electrical requirements of 3.7.3 at times specified, when tested in accordance with 4.8.5.

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3.8.6 Shock, bench-handling. The equipment shall show no evidence of physical damage except minor surface abrasion and shall meet the requirements of 3.7.3, when tested in accordance with 4.8.6.

3.8.7 Electromagnetic interference. The panoramic indicator shall comply with CE01, CE03, CS01, CS02, CS03, CS04, CS05, CS06, RE02 and RS03 interference and susceptibility requirements for Class 1C equipments of MIL-STD-461 when tested in accordance with 4.8.7.

3.9 Visual and mechanical. The equipment shall meet the requirements of MIL-STD-252 as specified in 4.9.

3.10 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or the equipment. Individual items shall not be hand picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristics, etc. (see 4.10).

3.11 Electrical preconditioning. Each equipment produced shall be subjected to the electrical preconditioning test of 4.11 after bounce preconditioning and prior to Group A inspection. The equipment shall be monitored for sweep rates of 3.7.3 and 4.7.3 during period of test.

3.12 Bounce preconditioning. Each equipment produced shall be subjected to the bounce preconditioning test of 4.6 prior to Group A inspection. After bounce preconditioning, the equipment shall not be repaired, aligned, cleaned or otherwise changed prior to performing electrical preconditioning and Group A inspection.

3.13 Workmanship. The equipment shall be manufactured and assembled in accordance with the applicable portions of MIL-P-11268 as follows:

General requirements for plastic material and parts.
Wiring and cabling including:

- Slack.
- Protection.
- Clearance.
- Splicing.
- Connectors, general.
- Grounding, general.

Flux and cleaning agents for soldering.
Process for soldering.
Cleaning of equipment.

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General process for securing of parts.
Process of welding.
Connectors, general.
Mechanical operation of controls.
Tropicalization of material.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the contract and specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection (Does not include preparation for delivery) (see 4.3).

(b) Inspection covered by subsidiary documents (see 4.4).

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.12).

4.3 First article inspection. This inspection shall be performed by the contractor and witnessed by a Government representative unless otherwise specified in the contract. It shall consist of Group A, B and C inspections (see Tables I, II and III respectively) on the three (3) samples of the completely assembled equipment. The first article samples shall be inspected in this order: Group A and B for all units; and designated units for specific Group C inspections. Designation of units to specific Group C inspections shall be determined by the Government production engineer. After completion of all Group C tests, conforming units must be reinspected and pass all Group A inspections.

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4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.5
Marking	3.6
Workmanship	3.13

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. Each unit produced shall be inspected by the contractor for conformance to all the Group A requirements and tests in Table I. Each lot shall be subject to Government verification and this inspection, including sampling, shall conform to Table I and the ordinary inspection procedures of MIL-STD-105.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in any order which is satisfactory to the Government.

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TABLE I. Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and mechanical	3.9	4.9	1 dphu*	4 dphu
Electrical				
Line positioning				
Horizontal	3.7.2.1	4.7.2.1		
Vertical	3.7.2.2	4.7.2.2		
Sweep rates				
1 Hz	3.7.3.1	4.7.3.1		
5 Hz	3.7.3.2	4.7.3.2		
30 Hz	3.7.3.3	4.7.3.3		
Variable sweep rate	3.7.3.4	4.7.3.4		
Hum	3.7.4	4.7.4		
Sensitivity				
Cross product sensitivity	3.7.5.1	4.7.5.1		
Direct input sensitivity	3.7.5.2	4.7.5.2		
Cross product reserve gain	3.7.6	4.7.6		
Calibration				
LIN amplitude scale	3.7.7.1	4.7.7.1		
LOG scale	3.7.7.2	4.7.7.2		
Frequency	3.7.7.3	4.7.7.3		
Sweep width			2.5 dphu for the group	**
Width	3.7.8.1	4.7.8.1		
Linearity	3.7.8.2	4.7.8.2		
Centering range				
Left	3.7.9.1	4.7.9.1		
Right	3.7.9.2	4.7.9.2		
Response curve	3.7.10	4.7.10		
Center frequency image rejection	3.7.11	4.7.11		
FM operations				
Multivibrator (2 Hz)	3.7.12.1	4.7.12.1		
Multivibrator (3 Hz)	3.7.12.2	4.7.12.2		
Distortion	3.7.12.3	4.7.12.3		
Intensifier	3.7.12.4	4.7.12.4		
Resolution capability	3.7.13	4.7.13		
Drop off at 30 Hz	3.7.14	4.7.14		
Video filter operation	3.7.15	4.7.15		

*Defects per hundred units
**All electrical tests are major

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4.5.2 Group B inspection. This inspection, including sampling, shall conform to Table II and to the special procedures for small sample inspection of MIL-STD-105. Group B inspection shall be performed on inspection lots that have passed Group A inspection, and on samples selected from units that have been subjected to and meet the Group A inspection.

4.5.2.1 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

TABLE II. Group B inspection

Inspection	Req. Para.	Test Para.	AQL
Interchangeability	3.10	4.10	4 dphu*
Electrical Operating voltage	3.7.1	4.7.1	1 dphu

* Defects per hundred units.

4.5.3 Group C inspection. This inspection shall consist of the inspection specified in Table III and shall be performed on sample units that have been subjected to and met Group A and B inspection. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for inspection.

4.5.3.1.1 Three (3) samples of the complete equipment shall be selected at random, one (1) sample for each Group 1 through Group 2 in Table III. The samples selected shall be at the start of the contract from the first quality conformance inspection lot. These samples shall constitute Group C requirement for first 100 units produced.

4.5.3.1.2 Thereafter, one (1) sample of the complete equipment selected at random for Group 2 in Table III from every 100 units and fraction thereof produced.

4.5.3.2 Evaluation of samples submitted to individual Group C inspection. After referenced samples in 4.5.3.1 have been subjected to an individual Group C inspection test and prior to subjection to the next Group C test (unless otherwise specified in the individual test) the sample shall be subject to and meet sweep rates of 3.7.3 and 4.7.3.

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4.5.3.3 Noncompliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Group A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.3.4 Order of inspection within Group C. Group C inspection shall be performed in any order which is satisfactory to the Government.

TABLE III. Group C inspection

Inspection	Req. Para.	Test Para.
Group 1		
Fungus	3.8.1	4.8.1
Group 2		
Altitude	3.8.2	4.8.2
Low temperature	3.8.3	4.8.3
High temperature	3.8.4	4.8.4
Humidity	3.8.5	4.8.5
Shock, bench-handling	3.8.6	4.8.6
Group 3		
Electromagnetic interference	3.8.7	4.8.7

4.5.4 Reinspection of conforming Group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are subjected to and pass Group A inspection.

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4.5.5 Disposition of Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order provided units meet requirements of 4.5.4.

4.6 Bounce preconditioning. Each equipment produced shall be placed in its formal operating position on the table of a package tester of suitable capacity as made by the L. A. B. Corporation, Skaneateles, New York with accessories as per Model 1000SC and SC-A-46439, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the equipment under test to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute and shall meet the requirements of 3.12.

4.7 Electrical tests.

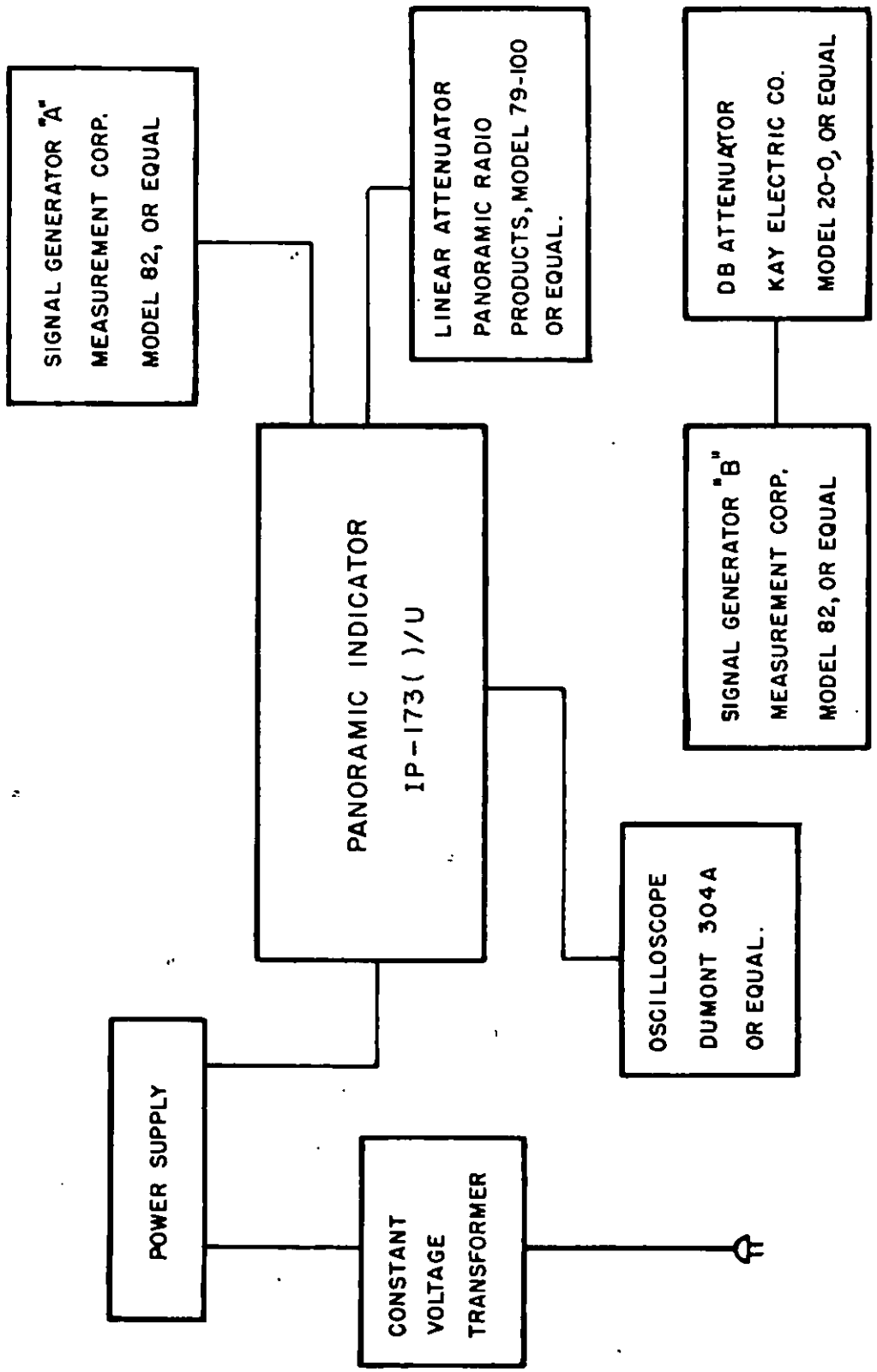
4.7.1 Operating voltage. Connect the equipment for normal operation with the constant voltage transformer between the indicator power supply, and a variable input power source. Vary the applied input between 95 Vac and 135 Vac while measuring the ac input voltage to the power supply. The voltage variation shall meet the requirements of 3.7.1.

4.7.2 Line positioning. With the equipment connected as shown in Figure I, set the front panel control as follows:

Center frequency to center position.
Sweep width fully clockwise.
Resolution fully clockwise.
Variable sweep rate to center position.
Sweep rate selector to 30.
Scale selector to LIN.
Gain fully clockwise.
Intensifier fully counter clockwise.
Video filter to Out.
Power to On.
Horizontal position for a centered trace.
Brilliance and focus as required for a sharp trace.

4.7.2.1 Horizontal. Apply a 5 MHz signal of sufficient amplitude to produce a pip approximately 3/4 of full scale deflection. Turn the sweep width control fully counter clockwise, the pip should spread until it becomes an elevated baseline. Set the center frequency control for maximum baseline rise and the sweep width fully clockwise; the pip should reappear.

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Set the horizontal position control at its maximum counter clockwise position, the pip shall be at least two divisions to the left of the center vertical grid line. Turn the horizontal position control fully clockwise, the pip shall move across the screen to at least the second division to the right of the center grid line. The sweep positions shall meet the requirements of 3.7.2.1.

4.7.2.2 Vertical. Recenter the pip. Reduce the 5 MHz input signal to zero input. Vary the vertical position control over its entire range. The baseline shall vary at least a half inch above and below the zero reference line of the linear scale. Set vertical position so that the baseline is at the zero reference of the linear scale. The sweep positions shall meet the requirements of 3.7.2.2.

4.7.3 Sweep rates. The sweep rate selector switch shall select sweep rates of 1, 5, and 30 Hz and a variable sweep rate. The variable sweep rate control shall adjust the sweep rate over the range of 25 to 35 Hz.

4.7.3.1 1 Hz. Set the sweep rate selector switch to "1". Select a point on the cathode ray tube(CRT) screen and, using a stop watch, determine the time it takes twenty sweeps to pass through this point. Repeat this test five times. The average time for twenty sweeps shall meet the requirements of 3.7.3.1.

4.7.3.2 5 Hz. Set the sweep width control fully counter clockwise, the sweep rate selector switch to "5" and the amplitude of the 5 MHz input signal to produce a 3/4 scale signal. Modulate the 5 MHz with a 60 Hz signal. The number of stationary sine waves that appear on the CRT shall meet the requirements of 3.7.3.2.

4.7.3.3 30 Hz. Set the sweep rate selector switch to "30". The number of stationary sine waves that appear on the CRT shall meet the requirements of 3.7.3.3. Remove the 60 Hz modulating selector.

4.7.3.4 Variable sweep rate. Set the sweep rate selector switch to "VAR," the variable sweep rate control fully counter clockwise, modulate the signal generator and adjust its output until one stable signal appears on the cathode ray tube screen. The modulating signal of the signal generator shall be 25 Hz or less. Repeat with the variable sweep rate control at its midpoint and fully clockwise. The modulating signal should be approximately 30 Hz and at least 35 Hz respectively, and shall meet the requirements of 3.7.3.4.

4.7.4 Hum. With no signal, the sweep rate selector switch at "5", and the gain control fully counter clockwise, the amplitude of any residual signals seen on the baseline shall meet the requirements of 3.7.4.

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4.7.5 Sensitivity.

4.7.5.1 Cross product sensitivity. With both signal generators connected as in Figure 1 and the sweep rate selector switch at "30". tune signal generator A to 15 MHz, 0.1 volt signal level, and signal generator B to 20 MHz, 0.05 volt signal level. Set the gain and sweep width controls fully clockwise and the scale selector switch to "LOG." The pip that appears on the CRT shall meet the requirements of 3.75.1.

4.7.5.2 Direct input sensitivity. With signal generator B off, signal generator A at 5 MHz and scale selector switch at "LIN", adjust signal generator A for a full scale deflection and measure the input signal. Repeat with signal generator A off and signal generator B tuned to 5 MHz, adjust signal generator B for a full scale deflection and measure the input signal. The measured input signals shall meet the requirements of 3.7.5.2.

4.7.6 Cross product reserve gain. With the scale selector switch at "LOG", set signal generator A to 15 MHz, 0.1 volt signal level, and signal generator B to 20 MHz. Adjust the output of signal generator B for a full scale pip, and record the output. Set the i.f. gain (on top of the chassis) fully clockwise. Reduce the output of signal generator B until a full scale pip is again seen on the screen. The output of signal generator B shall meet the requirements of 3.7.6. Reset signal generator B to the above recorded value and return the i.f. gain control to its original setting for a full scale pip.

4.7.7 Calibration. Set the controls as follows:

Scale selector switch to "LIN"
Signal generator A to 5 MHz.
Signal generator B to OFF.
DB attenuator to 20 dB.

4.7.7.1 LIN amplitude scale. Adjust the signal level until a full scale pip is obtained on the screen. Insert attenuation in nine steps, decreasing the input signal level to .9, .8, .7, .6, .5, .4, .3, .2 and .1 of its original value. The signal height at each step shall meet the requirements of 3.7.7.1

4.7.7.2 LOG scale. Set the linear attenuator to 1.0, the scale selector switch to "LOG" and the dB attenuator to "0 DB." Insert attenuator in ten steps, each step adding 4 dB. The signal height at each step shall meet the requirement of 3.7.7.2.

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4.7.7.3 Frequency. Set the sweep width to "0.5." Vary the signal generator a total of 500 kHz, (from 4.75 to 5.25 MHz) and increase the output until pips appear at the zero grid line, the +0.5 grid line, and the -0.5 grid line. The results of this test shall meet the requirements of 3.7.7.3.

4.7.8 Sweep width.

4.7.8.1 Width. With the sweep width control set to "1", the sweep width range shall meet the requirements of 3.7.8.1.

4.7.8.2 Linearity. Set the signal generator to 5 MHz, position the pip on the zero kHz grid line, and vary the frequency in increments of 100 kHz (from 5 MHz to 4.5 MHz and from 5 MHz to 5.5 MHz). When the pip is initially placed on the center grid line at a frequency of 5 MHz, the position of the pip shall meet the requirements of 3.7.8.2.

4.7.9 Centering range. The center frequency control range shall be at least 50 kHz. Set the scale selector switch to "LIN".

4.7.9.1 Left. Set the center frequency control fully counter clockwise. Tune the signal generator to 5 MHz. A pip shall appear and shall meet the requirements of 3.7,9.1.

4.7.9.2 Right. Set the center frequency control fully clockwise. A pip shall appear and shall meet the requirements of 3.7.9.2.

4.7.10 Response curve. Tune the signal generator to a 5 MHz signal and adjust the signal strength to produce a full scale pip. While maintaining a constant input level, vary the input frequency from 4.5 to 5.5 MHz. The pip height shall remain within 7 percent of the full scale value and shall meet the requirements of 3.7.10.

4.7.11 Center frequency image rejection. Set the gain control fully clockwise, the scale selector to "LIN" and inject a 5 MHz, 100 uV signal. Adjust the gain control for a 0.2 full scale pip. Increase the input frequency to 8 MHz and adjust the input signal to produce a 0.2 full scale pip. The input signal level shall be at least 15,000 uV and shall meet the requirements of 3.7.11.

4.7.12 FM operations.

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4.7.12.1 Multivibrator (2 Hz). Connect an oscilloscope to the multivibrator jack at the rear of the chassis and set the sweep rate selector switch to "30". The sine wave appearing on the oscilloscope screen shall meet the requirements of 3.7.12.1.

4.7.12.2 Multivibrator (3 Hz). Change the sweep rate selector switch to "5". The sine wave appearing on the oscilloscope shall meet the requirements of 3.7.12.2.

4.7.12.3 Distortion. Disconnect the oscilloscope. Change the sweep rate selector switch to "1", inject a 5 MHz, 200 Hz modulated signal. Adjust the sweep width and resolution controls (clockwise) for a distinct full scale center pip plus a full scale side pip. (Adjustment of the signal generator gain controls may be required to obtain these two pips as required). Adjust the resolution control until the difference in height between the peak of the center pip and the valley between the center and side pips is three screen divisions, maintaining a full scale center pip with the signal generator. Turn the sweep width control fully counter clockwise. Turn off the low frequency modulating signal. Adjust the input signal frequency until a maximum ripple appears on the baseline. The peak-to-peak ripple amplitude shall meet the requirements of 3.7.12.3.

4.7.12.4 Intensifier. Turn the sweep rate selector switch to "30". the sweep width and resolution controls fully clockwise, and inject a 5 MHz signal of sufficient strength to produce a full scale pip. Turn the brilliance control counter clockwise until the baseline disappears. Turn the intensifier control clockwise. The results of this test shall meet the requirements of 3.7.12.4.

4.7.13 Resolution capability. Adjust the brilliance control to its normal position, the sweep rate selector switch to "30". the intensifier control full counter clockwise, and inject a 5 MHz, 200 Hz modulated signal. Adjust the sweep width and resolution controls (counter clockwise) until a center pip, plus an adjacent pip of equal amplitude, appear on the cathode ray tube screen (adjustment of the signal generator gain controls may be required to obtain the required pips). Adjust the signal generator output for a full scale center pip. There shall be three calibration' divisions between the peak of the center pip and the valley between the center and side pips (at least 3 dB). The results of this test shall meet the requirements of 3.7.13.

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4.7.14 Drop off at 30 Hz. Set the sweep rate selector to "30", the sweep width fully clockwise, and the signal generator for a full scale center pip at 5 MHz. Turn the sweep width fully counter clockwise. The pip shall broaden until it becomes an elevated baseline. Adjust the center frequency control to maximum clockwise position. Adjust the signal generator output level until the baseline is behind the top calibration line of the cathode ray tube grid screen. Turn the sweep width control fully clockwise. The peak of the pip shall not drop more than 20 percent and shall meet the requirement of 3.7.14.

4.7.15 Video filter operation. Set the sweep width fully clockwise, and switch in the video filter. The amplitude of the pip shall meet the requirement of 3.7.15.

4.8 Service conditions. The Panoramic Indicator IP-173( )/U shall be tested in accordance with the following service conditions.

4.8.1 Fungus. The equipment in its normal operating configuration shall be subjected to 28 days of the fungus test specified in Procedure I, Method 508 of MIL-STD-810 and shall meet the requirements of 3.8.1.

4.8.2 Altitude. The equipment in its normal operating configuration shall be subjected to the altitude test specified in Procedure I, Method 500 of MIL-STD-810, with altitude in Step 3 held for 1 hour, and shall meet the requirements of 3.8.2.

4.8.3 Low temperature. The equipment in its normal operating configuration shall be subjected to the low temperature test specified in Procedure I, Method 502 of MIL-STD-810, with temperatures in Step 2 held for 2 hours and in Step 4 for 2 hours, and shall meet the requirements of 3.8.3.

4.8.4 High temperature. The equipment in its normal operating configuration shall be subjected to the high temperature test specified in Procedure I, Method 501 of MIL-STD-810, with temperature in Step 4 held for 2 hours, and shall meet the requirements of 3.8.4.

4.8.5 Humidity. The equipment in its normal operating configuration shall be subjected to the humidity test specified in Procedure II, Method 507 of MIL-STD-810, and shall meet the requirements of 3.8.5.

4.8.6 Shock, bench-handling. The equipment less the dust cover shall be positioned as for servicing on a solid two inch fir bench top and shall be subjected to the following:

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(a) Tilt up the equipment through an angle of 30 degrees, using one edge of the equipment as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axis, for a total of four drops on horizontal face.

(b) Repeat (a) with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the equipment could be placed practicably during servicing.

(c) The equipment shall meet the requirements of 3.8.6.

4.8.7 Electromagnetic interference. The panoramic indicator shall meet the requirements of 3.8.7. All test set-ups and procedures employed shall conform to the measurement techniques of MIL-STD-462.

4.9 Visual and mechanical. The equipment shall be examined for the defects listed in MIL-STD-252 and shall meet the requirements of 3.9.

4.10 Interchangeability. The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirements of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of case, cover and insertable assemblies, when such dimensions affect mating of parts.

(b) Location of hinges and fasteners on separable parts or assemblies which must mate, such as case, covers, and mountings.

(c) Location of switches, sockets, and fasteners.

(d) Location of cover and fasteners on carrying case which must mate.

(e) Location of panel mounting screw holes on basic unit for mating with case.

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4.11 Electrical preconditioning. Each equipment produced shall be placed in its normal operating position on a test table with room temperature at 77°F ±5°F, with specified operating voltage applied from a suitable power source. With all controls in the normal operating position, the equipment shall be turned on and remain on for a continuous 24 hour period. In case of parts failure, the equipment shall be repaired, recalibrated, and the electrical preconditioning test repeated and shall meet the requirements of 3.11. A failed item analysis report shall be submitted on each part failure with contents of report per 5.10.5 of MIL-STD-781.

4.12 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY.

5.1 Preservation, packaging, packing and marking shall be in accordance with MIL-I-55591. . . .

6. NOTES.

6.1 Intended use. The Panoramic Indicator IP-173( )/U is intended for use as a general purpose direct and general support equipment to measure amplitude and frequency of rf signals up to 1 GHz.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment (Levels A, B or C) (see section 5).

(c) If rough handling and functional tests are required.

(d) Quantity of first articles required.

(e) Place of final inspection.

(f) Technical literature required.

(g) Quantity of tools and running spare parts required.

(h) Marking for shipment and shipping containers.

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6.3 Nomenclature and type designation. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. The parentheses in the type designation will be deleted or replaced by a letter identifying the particular design for example: SG-12F/U or AN/URM-730.

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.3).

6.5 Temperature conversion. Temperature conversion from Fahrenheit to centigrade as listed in this specification are as follows:

160°F	=	71°C
125°F	=	51.5°C
77°F	=	25°C
-4°F	=	-20°C
-65°F	=	-53,9°C

6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodians:
Army - EL

Preparing activity
Army - EL

Project Number 6625-A353

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-I-55633(EL) INDICATOR, PANORAMIC IP-173( )/U

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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